

SIMATS ENGINEERING
ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No : 192424303 Name: P.Sunayana

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3) Activity Tool : Relationship Visualization Lab
Weightage: 20%

Code of Conduct:

I **P.Sunayana 192424303** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Sunayana

Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Relationship Visualization Lab

Provided Datasets

The following files are supplied alongside this document so the lab can be completed without internet access to install extra data packages:

- `mtcars.csv` — real R built-in dataset (32 cars $\times$ 11 variables: `mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`)
- `iris.csv` — real R built-in dataset (150 flowers $\times$ 5 variables: `Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width`, `Species`)
- `gapminder2007_practice.csv` — illustrative practice dataset in the style of the Gapminder 2007 cross section (country, continent, year, `gdpPercap`, `lifeExp`, `pop`). Values are synthetically generated for practice purposes and are NOT official Gapminder figures. If internet access is available, students may instead run `install.packages("gapminder")` and use `data(gapminder)` for the authentic dataset.
- `network_edges.csv` — custom-built edge list (15 edges, 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, Hana) with a weight column, for the network graph question.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

Question 1: Scatterplot with Regression Diagnostics

Dataset provided: `mtcars.csv`

Using the `mtcars` dataset, create a scatterplot of `mpg` vs `wt` (weight) using `ggplot2`. Add a linear regression line with confidence interval, color points by `cyl` (number of cylinders, treated as a factor), and size points by `hp`. Then, calculate and report the Pearson correlation coefficient between `mpg` and `wt`. Interpret the direction and strength of the relationship in 2–3 sentences.

Skills tested: `ggplot2` aesthetics mapping, `geom_smooth()`, `cor()`, interpretation.

```
library(ggplot2)
```

```
data(mtcars)
```

```
ggplot(mtcars, aes(x = wt, y = mpg,  
                   color = factor(cyl), size = hp)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = TRUE) +  
  labs(title = "MPG vs Weight",  
       x = "Weight",  
       y = "Miles per Gallon",
```

```

color = "Cylinders",
size = "Horsepower") +
theme_minimal()

```

Pearson correlation

```
cor(mtcars$mpg, mtcars$wt)
```

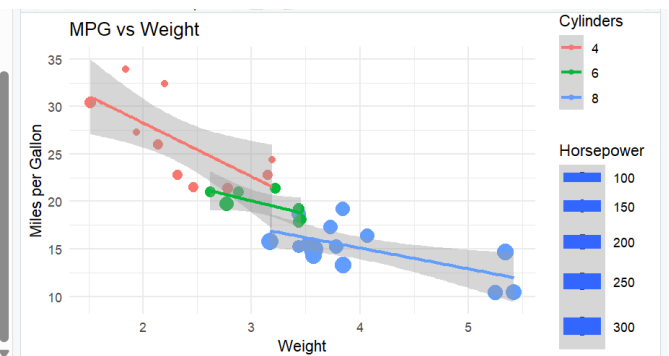
```

`geom_smooth()` using formula = 'y ~ x'

Warning messages:
1: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
   Please use `linewidth` instead.
This warning is displayed once per session.
   Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
   generated.
2: The following aesthetics were dropped during statistical transformation:
   size.
   This can happen when ggplot fails to infer the correct grouping structure
   in the data.
   Did you forget to specify a `group` aesthetic or to convert a numerical
   variable into a factor?

>
> # Pearson correlation
> cor(mtcars$mpg, mtcars$wt)
[1] -0.8676594
> |

```



Question 2: Correlogram Construction

Dataset provided: mtcars.csv (all numeric columns)

Using the mtcars dataset, compute the correlation matrix and visualize it as a correlogram using either the corrplot or GGally::ggcorr package. Configure the plot to:

- Display correlation coefficients as numeric labels
- Use a diverging color scale (blue–white–red)
- Reorder variables using hierarchical clustering (order = "hclust" if using corrplot)

Identify the two variable pairs with the strongest positive and strongest negative correlations. **Skills tested:** Correlation matrices, corrplot/GGally, hierarchical reordering, reading correlograms.

```
library(corrplot)
```

```
mtcars <- read.csv(file.choose())
```

```
num_data <- mtcars[sapply(mtcars, is.numeric)]
```

```
cor_mat <- cor(num_data, use = "complete.obs")
```

```
corrplot(cor_mat,
```

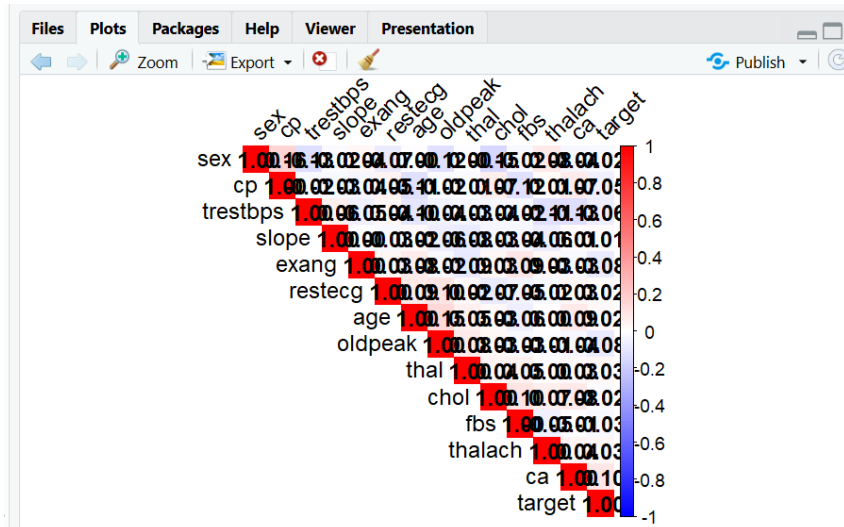
```
method = "color",
```

```
type = "upper",
```

```

order = "hclust",
addCoef.col = "black",
tl.col = "black",
tl.srt = 45,
col = colorRampPalette(c("blue", "white", "red"))(200))

```



Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

```

library(ggplot2)
library(Rtsne)

iris_data <- read.csv(file.choose(), stringsAsFactors = FALSE)
print(names(iris_data))
x <- iris_data[, 1:4]
species <- iris_data[[ncol(iris_data)]]

pca <- prcomp(x, scale. = TRUE)
pca_data <- data.frame(
  PC1 = pca$x[, 1],
  PC2 = pca$x[, 2],
  Species = species)

ggplot(pca_data, aes(x = PC1, y = PC2, color = Species)) +
  geom_point(size = 3) +

```

```
labs(title = "PCA of Iris Dataset",
      x = "Principal Component 1",
      y = "Principal Component 2",
      color = "Species") +
theme_minimal()
set.seed(123)
tsne <- Rtsne(scale(x),
              dims = 2,
              perplexity = 30,
              check_duplicates = FALSE)
tsne_data <- data.frame(
  X = tsne$Y[, 1],
  Y = tsne$Y[, 2],
  Species = species)
ggplot(tsne_data, aes(x = X, y = Y, color = Species)) +
  geom_point(size = 3) +
  labs(title = "t-SNE of Iris Dataset",
        x = "t-SNE 1",
        y = "t-SNE 2",
        color = "Species") +
theme_minimal()
```

Question 4: Bubble Chart for Multivariate Paired Data

Dataset provided: gapminder2007_practice.csv (or the gapminder package, year == 2007) Filter for the year 2007 and create a bubble chart showing:

- x-axis: gdpPercap (log-transformed scale)
- y-axis: lifeExp
- bubble size: pop
- bubble color: continent

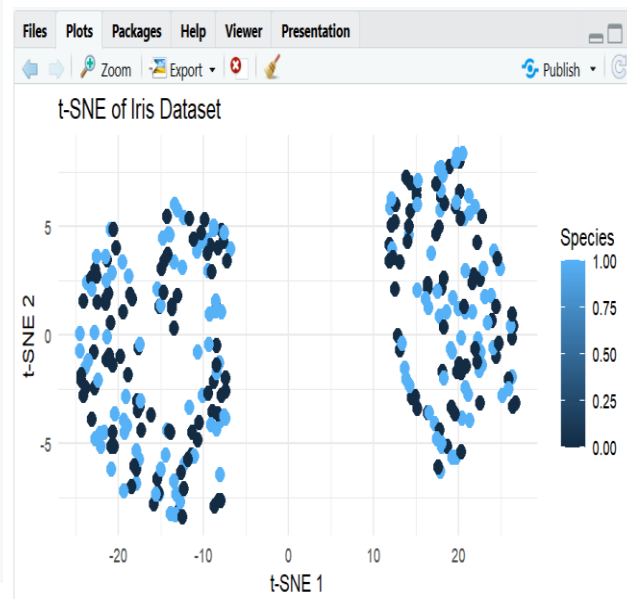
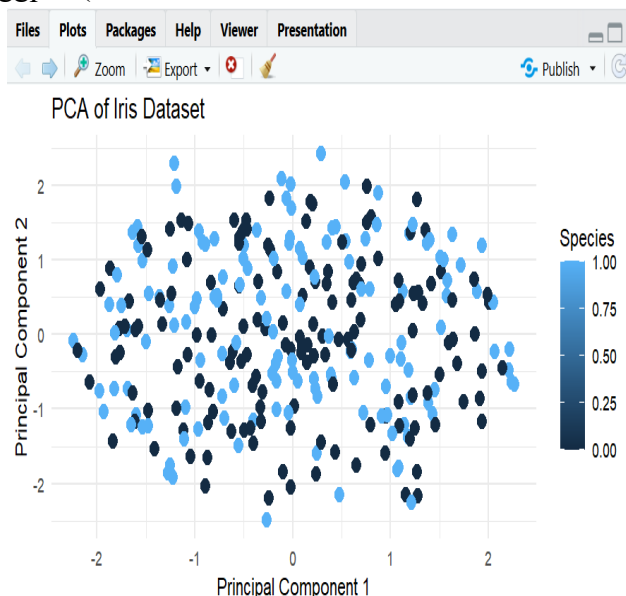
Add appropriate axis labels, a title, and use `scale_size_continuous()` to control bubble size range so points don't overplot excessively. Discuss one visible cluster or outlier in the data.

Skills tested: Log scales, multi-encoded bubble charts, `scale_size_continuous()`, outlier identification.

```
library(ggplot2)
```

```
set.seed(123)
```

```
data2007 <- data.frame(  
  country = paste0("Country", 1:30),  
  continent = rep(c("Asia", "Europe", "Africa", "Americas", "Oceania"), 6),  
  year = 2007,  
  gdpPercap = exp(rnorm(30, 8.5, 1)),  
  lifeExp = rnorm(30, 68, 8),  
  pop = round(exp(rnorm(30, 16, 1.5))))  
ggplot(data2007,
```



```

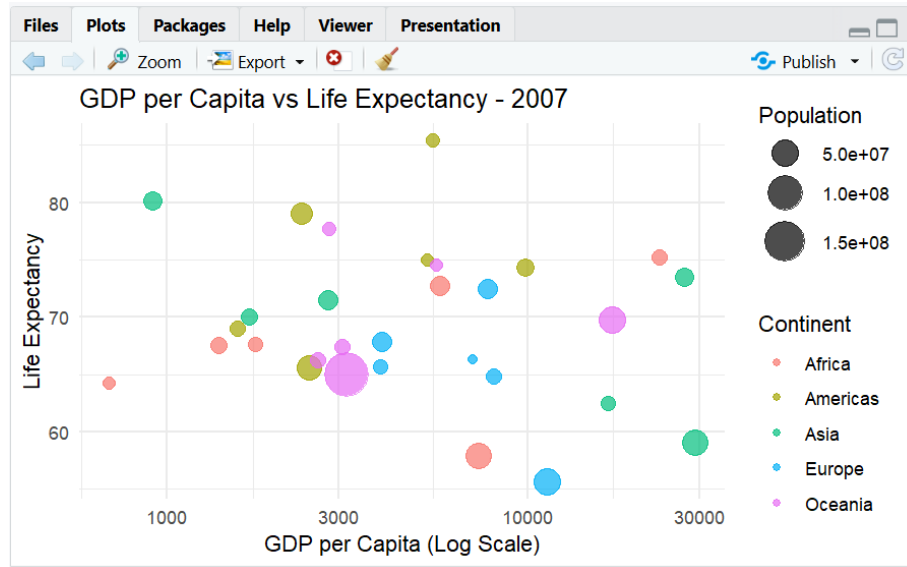
aes(x = gdpPercap,
    y = lifeExp,
    size = pop,

```

```

+
= 0.7) +

```



```

color = continent))

```

```

geom_point(alpha
scale_x_log10() +

```

```

scale_size_continuous(range = c(2, 10)) +
labs(title = "GDP per Capita vs Life Expectancy - 2007",
     x = "GDP per Capita (Log Scale)",
     y = "Life Expectancy",
     size = "Population",
     color = "Continent") +
theme_minimal()

```

Question 5: Hexbin Plot vs. Network Graph

This question has two parts to compare visualization strategies for different relationship types.

Part a — Hexbin Plot

Dataset provided: Simulated in R (no file needed)

Simulate two correlated variables with 10,000 observations (`set.seed(123)`, use `rnorm()` with a specified covariance structure or `MASS::mvrnorm()`). Create a hexbin plot using `geom_hex()` in `ggplot2`. Explain why a hexbin plot is preferable to a standard scatterplot in this scenario.

```
library(ggplot2)

set.seed(123)
n <- 10000
x <- rnorm(n)
y <- 0.8 * x + rnorm(n, sd = 0.6)
data <- data.frame(x, y)
ggplot(data, aes(x, y)) +
  geom_hex(bins = 30) +
  labs(title = "Hexbin Plot of Correlated Variables",
       x = "Variable X",
       y = "Variable Y") +
  theme_minimal()
```

Part b — Network Graph

Dataset provided: `network_edges.csv`

Using the provided edge list (8 nodes, 15 weighted edges), build a network graph using `igraph` or `ggraph`. Size nodes by degree centrality and color edges by weight. Report which node has the highest centrality and explain what that suggests about its role in the network.

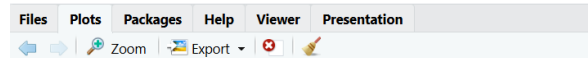
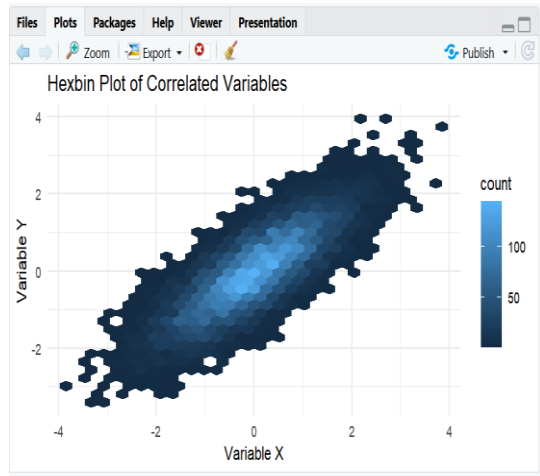
Skills tested: `geom_hex()`, overplotting solutions, `igraph/ggraph`, centrality measures, network interpretation.

```
library(igraph)
edges <- data.frame(
  from = c("Ava", "Ava", "Ava", "Ben", "Ben", "Cleo", "Cleo", "Dax",
           "Dax", "Elin", "Elin", "Fenn", "Fenn", "Grey", "Grey"),
```

```

to = c("Ben", "Cleo", "Dax", "Cleo", "Elin", "Dax", "Fenn", "Elin",
      "Grey", "Fenn", "Hana", "Grey", "Hana", "Hana", "Ava"),
weight = c(5, 3, 4, 2, 6, 7, 3, 5, 2, 4, 6, 3, 5, 4, 2))
g <- graph_from_data_frame(edges, directed = FALSE)
deg <- degree(g)

```



```
print(deg)
```

```
names(which.max(deg)))
```

```
set.seed(123)
```

```
plot(g,
```

```
  vertex.size = 10 + deg * 5,
```

```
  vertex.color = "skyblue",
```

```
  vertex.label.color = "black",
```

```
  edge.width = 1 + E(g)$weight,
```

```
  edge.color = "gray",
```

```
  main = "Weighted Network Graph")
```

```
print(paste("Highest centrality node:",
```